

Year 7 · Chapter 5B · Equations: answers (teacher copy)

5.10 Guess, check and improve (booklet p. 3)

Examples (model these live)

E1 Solve in your head: $x + 7 = 12$
 $5 + 7 = 12$ · $x = 5$

E2 Is $m = 4$ a solution of $3m + 2 = 15$?
 $3 \times 4 + 2 = 14$ · no: $14 \neq 15$

E3 Guess, check and improve: $6p + 5 = 77$
try 10: 65, too small · try 13: 83, too big · try 12: $77 \checkmark$, $p = 12$

Questions (38 on today's skill, 16 from earlier lessons)

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|---|--|---|---|
| 1 $y = 3$ | 19 $3 \times 8 + 7 = 31$ · no: $31 \neq 28$ | 31 try 11: 96 (too small) · try 14: 120 (too big) · try 13: 112 $\checkmark$, $x = 13$ | 42 try 11 and 12: $23 \cdot 12$ and 13 |
| 2 $x = 55$ | 20 $6 \times 11 + 16 = 82$: too small · $y = 13$ | 32 $3 \times 7 + 12 = 33$: too small · $n = 10$ | 43 $7 \times m \times n$ 5.03 |
| 3 yes | 21 $= 49 + 51 + 42 + 16 \cdot = 100 + 58 \cdot = 158$ 5.01 | 33 $7x$ 5.07 | 44 try 8: 72 · try 9: 90 $\checkmark$, the number is 9 |
| 4 $w = 9$ | 22 $x = 12$ | 34 $b = 7$ | 45 try 9 and 10: $19 \cdot 10$ and 11 |
| 5 $x = 48$ | 23 $2 \times 2 + 5 = 9$ · no: $9 \neq 11$ | 35 $= 4 \times (-3) + 9 = -3$ 5.05 | 46 try 3: 12 · try 4: 20 $\checkmark$, the number is 4 |
| 6 no (it gives 38) | 24 $= 7b \times 5n \cdot = 35bn \text{ cm}^2$ 5.08 | 36 try 3: 35 (too small) · try 7: 63 (too big) · try 6: 56 $\checkmark$, $x = 6$ | 47 2 4.06 |
| 7 $b = 7$ | 25 $7 \times 10 + 10 = 80$: too small · $m = 13$ | 37 try 7 and 8: $15 \cdot 8$ and 9 | 48 try 10: 48 (too small) · try 14: 60 (too big) · try 13: 57 $\checkmark$, $a = 13$ |
| 8 $n = 4$ | 26 $6 \times 5 + 12 = 42$: too small · $a = 6$ | 38 50% of 180 $\cdot = 90$ marks 4.11 | 49 $c - 4$ 5.04 |
| 9 9 5.05 | 27 $y = 5$ | 39 try 6: 42 · try 7: 56 $\checkmark$, the number is 7 | 50 $n = 12$ |
| 10 $n = 18$ | 28 $\frac{22}{5}$ 4.01 | 40 9 5.09 | 51 try 6 and 7: $13 \cdot 7$ and 8 |
| 11 yes | 29 $6 \times 6 + 5 = 41$ · no: $41 \neq 35$ | 41 try 6: 30 (too small) · try 10: 46 (too big) · try 9: 42 $\checkmark$, $x = 9$ | 52 no 5.07 |
| 12 $w = 1$ | 30 $5 \times 36 = 180 \cdot 150 + 30 = 180$ 5.02 | | 53 try 2: 6 · try 3: 12 $\checkmark$, the number is 3 |
| 13 0.35 4.10 | | | 54 $= 9 \times 6 \cdot = 54$ 5.05 |
| 14 $2 \times 8 + 1 = 17$ · yes: $17 = 17$ | | | |
| 15 $t = 72$ | | | |
| 16 yes | | | |
| 17 n 5.09 | | | |
| 18 $p = 3$ | | | |

5.11 Solving one-step equations (booklet p. 5)

Examples (model these live)

E1 Solve $x + 9 = 23$.
 $x + 9 - 9 = 23 - 9$ · $x = 14$

E2 Solve $4m = 36$.
 $4m \div 4 = 36 \div 4$ · $m = 9$

E3 Solve $\frac{p}{5} = 7$.
 $\frac{p}{5} \times 5 = 7 \times 5$ · $p = 35$

Questions (40 on today's skill, 17 from earlier lessons)

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|--|--|--|--|
| 1 $t = 19$ | 18 $t = 6$ | 32 $m = 23 \div 2 \cdot m = 11\frac{1}{2}$ | 44 $n = 29 \div 2 \cdot n = 14\frac{1}{2}$ |
| 2 $n = 31$ | 19 $4m \div 4 = 36 \div 4 \cdot m = 9$ | 33 $\frac{a}{5} \times 5 = 8 \times 5 \cdot a = 40$ | 45 $9 \times n \times x$ 5.03 |
| 3 $\times 5$ | 20 $\frac{x}{8} \times 8 = 8 \times 8 \cdot x = 64$ | 34 no (it gives 15) 5.10 | 46 $8w \div 8 = -32 \div 8 \cdot w = -4$ |
| 4 $k = 22$ | 21 $= 35 + 65 + 47 + 40 \cdot = 100 + 87 \cdot = 187$ 5.01 | 35 $n = 0 - 5 \cdot n = -5$ | 47 $4 \times 9 + 10 = 46$ · yes: $46 = 46$ 5.10 |
| 5 $n = 24$ | 22 $y = 3 - 4 \cdot y = -1$ | 36 numbers: $16 \div 4 = 4$; letters: $a \cdot = 4a$ 5.09 | 48 $5t \div 5 = -15 \div 5 \cdot t = -3$ |
| 6 -7 | 23 $8a \div 8 = 24 \div 8 \cdot a = 3$ | 37 $7k \div 7 = 77 \div 7 \cdot k = 11$ | 49 $34 - 26 = t \cdot t = 8$ |
| 7 $y = 21$ | 24 $= 5b \times 3n \cdot = 15bn \text{ cm}^2$ 5.08 | 38 $p = 25 \div 9 \cdot p = 2\frac{7}{9}$ | 50 50 3.03 |
| 8 $x = 32$ | 25 $\frac{x}{4} \times 4 = 12 \times 4 \cdot x = 48$ | 39 $3a \div 3 = -9 \div 3 \cdot a = -3$ | 51 Subtracted 2 instead of dividing by 2. $t = 3$. |
| 9 $t = 30$ 5.10 | 26 $w = 8 - 12 \cdot w = -4$ | 40 $= 200 \times 6 + 2 \times 6 \cdot = 1200 + 12 \cdot = 1212$ 3.02 | 52 $c - 3$ 5.04 |
| 10 $p = 49$ | 27 $\frac{w}{6} \times 6 = 2 \times 6 \cdot w = 12$ | 41 $56 - 47 = k \cdot k = 9$ | 53 $y = 11 \div 4 \cdot y = 2\frac{3}{4}$ |
| 11 $+4$ | 28 $t = 5 - 10 \cdot t = -5$ | 42 $4c^2$ 5.08 | 54 $w = 4 - 11 \cdot w = -7$ |
| 12 $a = 12$ | 29 $\frac{5}{9}$ 4.03 | 43 Subtracted 6 instead of dividing by 6. $a = 2$. | 55 $4m$ 5.09 |
| 13 49 3.07 | 30 $8k \div 8 = 24 \div 8 \cdot k = 3$ | | 56 $8p \div 8 = -40 \div 8 \cdot p = -5$ |
| 14 $3m \div 3 = 12 \div 3 \cdot m = 4$ | 31 $5 \times 53 = 265 \cdot 250 + 15 = 265$ 5.02 | | 57 $= 5 \times 3 \times mx \times m \cdot = 15m^2x$ 5.08 |
| 15 $a = 18$ | | | |
| 16 $\div 3$ | | | |
| 17 x 5.09 | | | |

5.12 Solving two-step equations (booklet p. 7)

Examples (model these live)

E1 Solve $3x + 5 = 26$.
 $3x = 26 - 5 = 21$ · $x = 21 \div 3 = 7$

E2 Solve $5m - 4 = 31$.
 $5m = 31 + 4 = 35$ · $m = 35 \div 5 = 7$

E3 Solve $2(a + 3) = 18$.
 $a + 3 = 18 \div 2 = 9$ · $a = 9 - 3 = 6$

Questions (40 on today's skill, 17 from earlier lessons)

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|--|---|--|--|
| 1 - 9 from both sides | 19 $8x = 39 - 15 = 24 \cdot x = 24 \div 8 = 3$ | 31 $9 \times 28 = 252 \cdot 180 + 72 = 252$ 5.02 | 44 $6b = -50 - 4 = -54 \cdot b = -54 \div 6 = -9$ |
| 2 $m = 3$ | 20 $\frac{t}{2} = 6 \cdot t = 6 \times 2 = 12$ | 32 $3a = 1 - 10 = -9 \cdot a = -9 \div 3 = -3$ | 45 $3 \times c \times x$ 5.03 |
| 3 no | 21 $= 18 + 82 + 83 + 23 \cdot = 100 + 106 \cdot = 206$ 5.01 | 33 $\frac{m}{6} = 8 \cdot m = 8 \times 6 = 48$ | 46 $8b = 14 \cdot b = 1\frac{6}{8}$ |
| 4 + 3 to both sides | 22 $n + 8 = 20 \div 2 = 10 \cdot n = 10 - 8 = 2$ | 34 + 4 5.11 | 47 $3a \div 3 = 33 \div 3 \cdot a = 11$ 5.11 |
| 5 $y = 5$ | 23 $5y = 46 - 16 = 30 \cdot y = 30 \div 5 = 6$ | 35 $n + 8 = 60 \div 4 = 15 \cdot n = 15 - 8 = 7$ | 48 $3m = 7 \cdot m = 2\frac{1}{3}$ |
| 6 no | 24 $20 \div 5 = 4 \cdot = 4n$ 5.09 | 36 $a = 7$ 5.10 | 49 $x - 7 = -5 \cdot x = -5 + 7 = 2$ |
| 7 + 8 to both sides | 25 $\frac{m}{4} = 4 \cdot m = 4 \times 4 = 16$ | 37 $8n = 18 + 14 = 32 \cdot n = 32 \div 8 = 4$ | 50 35.8 4.11 |
| 8 $y = 1$ | 26 $p + 3 = 48 \div 4 = 12 \cdot p = 12 - 3 = 9$ | 38 $6x = -29 - 7 = -36 \cdot x = -36 \div 6 = -6$ | 51 $4p = 20$ is right, but then divide by 4, do not subtract it: $p = 5$. |
| 9 $p = 27$ 5.11 | 27 $\frac{a}{2} = 3 \cdot a = 3 \times 2 = 6$ | 39 $9x = 12 \cdot x = 1\frac{3}{9}$ | 52 $c - 10$ 5.04 |
| 10 $a = 2$ | 28 $y + 7 = 96 \div 6 = 16 \cdot y = 16 - 7 = 9$ | 40 $= 400 \times 4 + 12 \times 4 \cdot = 1600 + 48 \cdot = 1648$ 3.02 | 53 $3y = 5 - 11 = -6 \cdot y = -6 \div 3 = -2$ |
| 11 no | 29 $\frac{7}{9}$ 4.06 | 41 $t - 4 = 3 \cdot t = 3 + 4 = 7$ | 54 $k + 9 = 105 \div 5 = 21 \cdot k = 21 - 9 = 12$ |
| 12 - 13 from both sides | 30 $7m = 59 + 18 = 77 \cdot m = 77 \div 7 = 11$ | 42 4 5.09 | 55 $y = 9$ 5.10 |
| 13 $\frac{5}{18}$ 4.12 | | 43 $6t = 18$ is right, but then divide by 6, do not subtract it: $t = 3$. | 56 $8t = 44 \cdot t = 5\frac{4}{8}$ |
| 14 $9m = 43 + 20 = 63 \cdot m = 63 \div 9 = 7$ | | | 57 $= \frac{54ct}{6t} \cdot = 9c$ 5.09 |
| 15 $a = 3$ | | | |
| 16 yes | | | |
| 17 $x = 16$ 5.10 | | | |
| 18 - 9 from both sides | | | |

5.13 Equations and problem solving (booklet p. 9)

Examples (model these live)

E1 A number times 4, plus 7, is 43. Find the number.

$$4n + 7 = 43 \cdot 4n = 36, n = 9$$

E2 6 equal packs hold 72 pencils. How many pencils are in each pack?

$$6p = 72 \cdot p = 12: 12 \text{ pencils}$$

E3 Two angles of a triangle are 50° and 65° . Find the third angle.

$$50 + 65 + x = 180 \cdot x = 65: 65^\circ$$

Questions (35 on today's skill, 14 from earlier lessons)

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|-------------------------------|--|---|--|
| 1 $n + 4 = 19$ | 17 $b - 7$ 5.04 | 29 $5 \times 53 = 265 \cdot 250 + 15 = 265$ 5.02 | 39 $25 + 25h = 75 \cdot 25h = 50 \cdot h = 2: 2 \text{ hours}$ |
| 2 12 | 18 20 | 30 $25 + 25h = 125 \cdot 25h = 100 \cdot h = 4: 4 \text{ hours}$ | 40 $4 \times n \times p$ 5.03 |
| 3 $5n = 20$ | 19 $5n + 9 = 59 \cdot n = 10$ | 31 $3p = 87 \cdot p = 29: 29 \text{ pencils}$ | 41 $w + w + 8 + w + w + 8 = 56 \cdot 4w + 16 = 56 \cdot w = 10: 10 \text{ cm}$ |
| 4 11 | 20 $7p = 203 \cdot p = 29: 29 \text{ pencils}$ | 32 $t = 18$ 5.10 | 42 $s + s + 6 = 20 \cdot 2s = 14 \cdot s = 7: \text{Sam is } 7$ |
| 5 $n - 3 = 13$ | 21 $= 22 + 78 + 46 + 11 \cdot = 100 + 57 \cdot = 157$ 5.01 | 33 $30 + 49h = 275 \cdot 49h = 245 \cdot h = 5: 5 \text{ hours}$ | 43 XXXV 3.06 |
| 6 9 | 22 $25 + 51 + x = 180 \cdot x = 104: 104^\circ$ | 34 $w + w + 2 + w + w + 2 = 64 \cdot 4w + 4 = 64 \cdot w = 15: 15 \text{ cm}$ | 44 $n + n + 1 + n + 2 = 36 \cdot 3n + 3 = 36, n = 11 \cdot 11, 12, 13$ |
| 7 $n \div 8 = 7$ | 23 $2n + 3 = 37 \cdot n = 17$ | 35 $\times 100: 80\% \cdot \frac{80}{100} = \frac{4}{5}$ 4.10 | 45 25 5.05 |
| 8 17 | 24 $\frac{k}{8} \times 8 = 7 \times 8 \cdot k = 56$ 5.11 | 36 $s + s + 6 = 18 \cdot 2s = 12 \cdot s = 6: \text{Sam is } 6$ | 46 $40 + 48h = 136 \cdot 48h = 96 \cdot h = 2: 2 \text{ hours}$ |
| 9 $k = 4$ 5.12 | 25 $7p = 56 \cdot p = 8: 8 \text{ pencils}$ | 37 $6a$ 5.04 | 47 $47 + 20 + x = 180 \cdot x = 113: 113^\circ$ |
| 10 17 | 26 $68 + 70 + x = 180 \cdot x = 42: 42^\circ$ | 38 $n + n + 1 + n + 2 = 39 \cdot 3n + 3 = 39, n = 12 \cdot 12, 13, 14$ | 48 no (it gives 45) 5.10 |
| 11 $n + 4 = 12$ | 27 $\frac{12}{27} \cdot = \frac{4}{9}$ 4.08 | | 49 $w + w + 7 + w + w + 7 = 54 \cdot 4w + 14 = 54 \cdot w = 10: 10 \text{ cm}$ |
| 12 $2n + 6 = 44 \cdot n = 19$ | 28 $3n - 5 = 31 \cdot n = 12$ | | |
| 13 $\frac{7}{2}$ 4.01 | | | |
| 14 15 | | | |
| 15 $7n = 112$ | | | |
| 16 $3n - 4 = 50 \cdot n = 18$ | | | |